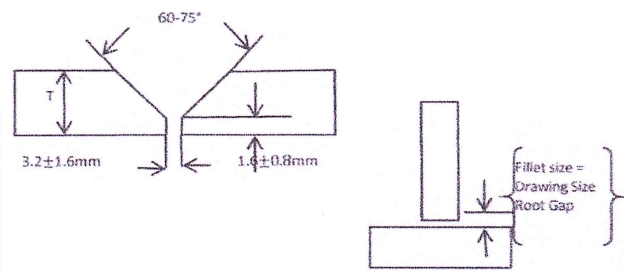


Welding Procedure Specification

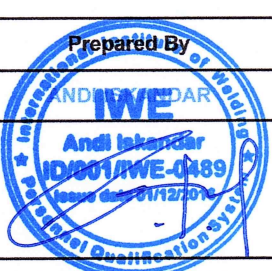
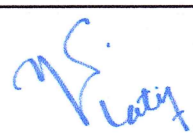
(Section IX, ASME Boiler and Pressure Vessel Code, Edition 2023)

WPS NO. :	WPS-PMC-ASME-1.1-GS-003	Rev. :	0	Date :	7-Sep-24
Supporting PQR No.(s) :	PQR-PMC-ASME-1.1-GS-003				
Welding Process(es) :	GTAW	SMAW	Type(s) :	MANUAL	
Applicable Code :	ASME BPVC IX 2023				

JOINTS (QW-402) Joint Design : <u>Groove&Fillet</u> Root Spacing : <u>See Joint Detail</u> Backing : ☐ Yes ☒ No Backing Material : <u>N/A</u> Other : <u>N/A</u>	Joint Detail 
---	---

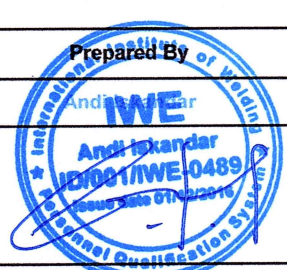
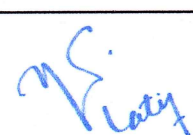
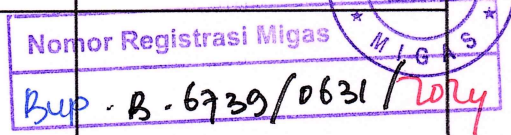
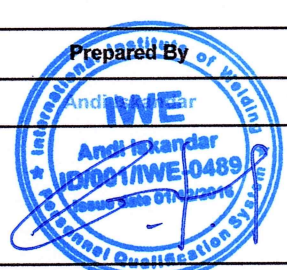
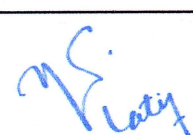
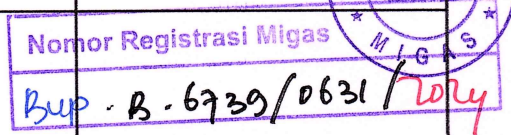
BASE METAL (QW-403)					
P. No. :	1	Group No. :	1&2	to P. No. :	1
Specification Type & Grade :		SA106 Gr. B or Equivalent		to SA 106 Gr. B or Equivalent	
Thickness Range	Groove :		5 mm - 43,9 mm	Fillet :	Unlimited
Maximum Pass Thickness ≤ 1/2 in. (13mm)			☒ Yes	☐ No	
Other	N/A				

FILLER METALS (QW-404)					
Process :	GTAW		SMAW		
Spec No. (SFA) :	5.18		5.1		
AWS. No. (Class) :	ER70S-G		E7018		
F. No. :	6		4		
A. No. :	1		1		
Size of Filler Metals :	See Below Table		See Below Table		
Weld Metal Thickness Range					
Groove :	Max. 12 mm		Max. 31,9 mm		
Fillet :	Unlimited		Unlimited		
Manufacturer :	-		-		
Brand :	-		-		
Other :	N/A		N/A		

Approvals	Prepared By	Checked By	Approved By
Name	ANDI INKANDAR IWE	LATIF	MIGAS IM
Signature	 ID/001/IWE-0489 Issue Date: 01/12/2024		Nomor Registrasi Migas* BUP-B-67309/0631/2024
Date	September 07, 2024	September 07, 2024	

Welding Procedure Specification

(Section IX, ASME Boiler and Pressure Vessel Code, Edition 2023)

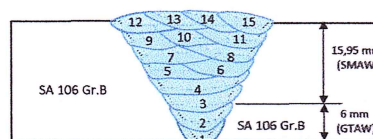
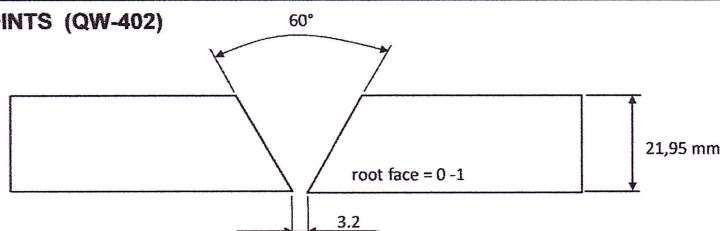
POSITION (QW-405) Position of Groove : <u>ALL</u> Welding Progression : ☒ Up ☐ Down Position of Fillet : <u>All</u> Other : <u>N/A</u>				POST WELD HEAT TREATMENT (QW-407) Temperature Range : <u>N/A</u> Time Range : <u>N/A</u> Other : <u>N/A</u>																																													
PREHEAT (QW-406) Preheat Preheat Temp. Min. : <u>10 °C</u> Interpass Temp, Max : <u>250 °C</u> Preheat Maintenance : <u>N/A</u> Other : <u>N/A</u>				GAS (QW-408) <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th colspan="3">Percent composition</th> </tr> <tr> <th></th> <th>Gas(es)</th> <th>(Mixture)</th> <th>Flowrate</th> </tr> </thead> <tbody> <tr> <td>Shielding</td> <td><u>Argon</u></td> <td><u>99.9%</u></td> <td><u>12-25 L/Min.</u></td> </tr> <tr> <td>Trailing</td> <td><u>N/A</u></td> <td><u>N/A</u></td> <td><u>N/A</u></td> </tr> <tr> <td>Backing</td> <td><u>N/A</u></td> <td><u>N/A</u></td> <td><u>N/A</u></td> </tr> </tbody> </table>				Percent composition				Gas(es)	(Mixture)	Flowrate	Shielding	<u>Argon</u>	<u>99.9%</u>	<u>12-25 L/Min.</u>	Trailing	<u>N/A</u>	<u>N/A</u>	<u>N/A</u>	Backing	<u>N/A</u>	<u>N/A</u>	<u>N/A</u>																							
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ELECTRICAL CHARACTERISTICS (QW-409) Process : <u>GTAW</u> <u>SMAW</u> Current & Polarity : <u>See Below Table</u> Amps (Range) : <u>See Below Table</u> Vols (Range) : <u>See Below Table</u> Tungsten Electrode Size and Type : <u>EW Th-2 / 2.4 mm Dia. (GTAW)</u> Electrode Wire Feed Speec Range : <u>N/A</u> Other : <u>N/A</u>																																																	
TECHNIQUE (QW-410) String or Weave Bead : <u>Both</u> Orifice, Nozzle, or Gas Cup Size : <u>6-12mm</u> Initial & Interpass Cleaning : <u>Wire Brush & Grinding</u> Method of Back Gouging : <u>N/A</u> Oscillation : <u>N/A</u> Contact Tube to Work Distance : <u>N/A</u> Multiple or single Pass : ☒ MULTI ☒ SINGLE Multiple or single Electrodes : ☐ MULTI ☒ SINGLE Peening : <u>N/A</u> Other : <u>N/A</u>																																																	
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th rowspan="2">Weld Pass(es)</th> <th rowspan="2">Process</th> <th colspan="2">Filler Metal</th> <th rowspan="2">Current Type and Polarity</th> <th rowspan="2">Amps (Range)</th> <th rowspan="2">Wire Feed Speed (Range)</th> <th rowspan="2">Volts (Range)</th> <th rowspan="2">Travel Speed (Range) cm/min</th> <th rowspan="2">Remarks</th> </tr> <tr> <th>Classifi-cation</th> <th>Dia. (mm)</th> </tr> </thead> <tbody> <tr> <td>Root/Hot</td> <td>GTAW</td> <td>ER70S-G</td> <td>2,4</td> <td>DCEN</td> <td>80-115</td> <td>N/A</td> <td>9 - 15</td> <td>3 - 10</td> <td></td> </tr> <tr> <td>Fill/Cap</td> <td>SMAW</td> <td>E7018</td> <td>2,6, 3.2, 4.0</td> <td>DCEP</td> <td>90-200</td> <td>N/A</td> <td>20 - 28</td> <td>5 - 13</td> <td></td> </tr> <tr> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> </tr> </tbody> </table>								Weld Pass(es)	Process	Filler Metal		Current Type and Polarity	Amps (Range)	Wire Feed Speed (Range)	Volts (Range)	Travel Speed (Range) cm/min	Remarks	Classifi-cation	Dia. (mm)	Root/Hot	GTAW	ER70S-G	2,4	DCEN	80-115	N/A	9 - 15	3 - 10		Fill/Cap	SMAW	E7018	2,6, 3.2, 4.0	DCEP	90-200	N/A	20 - 28	5 - 13											
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Remarks :																																																	
Approvals		Prepared By		Checked By		Approved By																																											
Name		Andika Iqbal 		LATIF 		MIGAS 																																											
Signature																																																	
Date		September 07, 2024		September 07, 2024																																													

PROCEDURE QUALIFICATION RECORDS (PQR)

(See QW-200.1 Section IX, ASME Boiler and Pressure Vessel Code, Edition 2023)

Company Name : **PT. PERTAMINA MAINTENANCE & CONSTRUCTION** Sub Contractor Name _____
 Procedure Qualification Record No. : **PQR-PMC-ASME-1.1-GS-003** Date **September 07, 2024**
 WPS No. : **WPS-PMC-ASME-1.1-GS-003** Date **September 07, 2024**
 Welding Process (es) **GTAW SMAW**
 Types (Manual, Automatic, Semi-Auto) **Manual**

JOINTS (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal and process used)

BASE METALS (QW-403)

Material Spec. **SA 106**
 Type/Grade, or UNS Number **Gr. B**
 P. No. **1** Group No. **1** to P-No. **1** Group No. **1**
 Thickness of Test Coupon **21,95 mm**
 Diameter of Test Coupon **6" (OD 168,3 mm)**
 Maximum Pass Thickness **2,5 mm**
 Other _____

POSTWELD HEAT TREATMENT (QW-407)

Temperature **NA**
 Time **NA**
 Other **NA**

GAS (QW-408)

	Gas (es)	(Mixture)	Flow Rate
Shielding	Argon	99,99%	15 L/min.
Trailing	NA		
Backing	NA		
Other	NA		

FILLER METALS (QW-404)

	GTAW (Root/Hot)	SMAW (Fill/Cap)
SFA Specification	5.18	5.1
AWS Classification	ER70S-6	E7018
Filler Metal F- No.	6	4
Weld Metal Analysis A-No.	1	1
Size of Filler Metal	2.4 mm	3.2, 4.0 mm
Filler Metal Product Form	Solid	NA
Supplemental Filler Metal	NA	NA
Electrode Flux Classification	NA	NA
Flux Type	NA	NA
Flux Trade Name	NA	NA
Weld Metal Thickness	6 mm	15,95
Other		

ELECTRICAL CHARACTERISTICS (QW-409)

Current **DC**
 Polarity **See Table**
 Amps. **See table** Volts **See Table**
 Tungsten Electrode Size **Ø 2.4 mm**
 Mode of Metal Transfer for GMAW (FCAW) **NA**
 Heat Input **NA**
 Other **NA**

POSITION (QW-405)

Position of Groove **6G**
 Weld Progression (Uphill, Downhill) **Uphill**
 Other **NA**

PREHEAT (QW-406)

Preheat Temp. **31 °C**
 Interpass Temp. **Maximum 243 °C**
 Other **NA**

TECHNIQUE (QW-410)

Travel Speed **See Table**
 String or Weave Bead **Weave**
 Oscillation **NA**
 Multipass or Single Pass (per side) _____
 Single or Multiple Electrodes _____
 Other **NA**

Nomor Registrasi Migas

BUF - B. 6735/0631/2024

PQR No. : PQR-PMC-ASME-1.1-GS-003

Welding Parameter

Weld Layers	Process	Filler Metal		Current		Voltage (V)	Travel Speed (mm/min)	Other
		Class	Dia. (mm)	Polarity	Ampere (A)			
1	GTAW	ER70S-6	2.4	DCEN	90	10	33,51	
2	GTAW	ER70S-6	2.4	DCEN	110	11	53,57	
3	SMAW	E7018	4.0	DCEP	125	25	69,31	
4	SMAW	E7018	4.0	DCEP	125	25	66,53	
5	SMAW	E7018	4.0	DCEP	125	25	75,37	
6	SMAW	E7018	4.0	DCEP	125	25	72,35	
7	SMAW	E7018	4.0	DCEP	125	25	69,31	
8	SMAW	E7018	4.0	DCEP	125	25	74,73	
9	SMAW	E7018	3.2	DCEP	120	24	71,63	
10	SMAW	E7018	3.2	DCEP	120	24	68,55	
11	SMAW	E7018	3.2	DCEP	120	24	76,27	
12	SMAW	E7018	3.2	DCEP	113	24	79,81	
13	SMAW	E7018	3.2	DCEP	113	24	75,19	
14	SMAW	E7018	3.2	DCEP	113	24	77,37	
15	SMAW	E7018	3.2	DCEP	113	24	81,35	

Tensile Test (QW-150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (kN)	Ultimate Unit Stress (N/mm ²)	Type of Failure and Location
T1	19,48	21,74	423,5	217,06	513	Ductile (Base Metal)
T2	19,51	21,76	424,54	216,31	510	Ductile (Base Metal)

Guided - Bend Test (QW-160)

Type and Figure No.	Result
Side Bend (SB1)	No Open Discontinuity was observed
Side Bend (SB2)	No Open Discontinuity was observed
Side Bend (SB3)	No Open Discontinuity was observed
Side Bend (SB4)	No Open Discontinuity was observed

Toughness Tests (QW-170)

Specimen No.	Notch Location	Specimen Size	Test Temperature	Impact Values			Drop Weight Break (Y/N)
				ft-lb or J	% Shear	Mils (in.) or mm	
NA							

 Comments : NA

Fillet - Weld Test (QW-180)

 Result - Satisfactory : Yes NA No. NA Penetration into parent metal : Yes NA No. NA

 Macro - Result NA

Other Tests

 Type of test NA

 Deposit Analysis NA

 Radiographic Result Accepted

 Report No. RPD/PERTAMC/RT/007

 Other NA

 Welder's Name Fadli

 Clock No. -

 Stamp No. -

 Test conducted by : PT. Hi-Test

 Laboratory Test No. HT/JKT/0824/0202

 Date August 30, 2024

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Boiler and Pressure Vessel Code.

Prepared By,

Reviewed By,

Approved By,

MIGAS

Nomor Registrasi Migas

Bup - B-6739/0631/2024

Date : September 07, 2024

Date : September 07, 2024

Date :



PT. RETSA PUTRA DYME
NDT TECHNICAL & INSPECTION SERVICE



HEAD OFFICE : Jl. Moch Ramdhan No.117 Blok Kapling Lebak, Kel. Sitiwinangun, Kecamatan Jamblang,
Kab. Cirebon Prov. Jawa Barat 45157
Email : retsaputradyme@gmail.com

RADIOGRAPHIC EXAMINATION REPORT

Client : PT. PERTAMC		Report No. : RPD/PERTAMC/RT/007		Radiographic Technique Sketch																
Project : PQR		Date : 24-08-2024		☐ DWID ELLIPSE ☐ SWSI PIPE																
Location : CILEGON		Procedure No. : -																		
Job / Contract No. :		Accept. Code : ASME IX																		
Type Material : SA 106 Gr.B	Radiation : Ir. 192	Penetrameter : ASTM IB																		
Thickness Mat'l : 21,95 mm	Focal Spot : 2 x 3 mm	Density : 2-4		☐ DWSI CONTACT ☐ SWSI PLATE																
Pipe ☒ Plate ☐ : 6 in/mm XXS	Activity : 20 Ci	Un. Geometry : 0,5																		
Weld Process : ☒ SMAW ☒ GTAW	SFD : 6 Inch	Sensitivity : 2%		☐ DWSI SUPERIMPOSED ☐ SWSI PANORAMIC																
☐ SAW ☐ FCAW	Exposure Time : 2 mnt	Screen Type : Pb 0,125																		
☐ GMAW ☐ Other	Placement IQI : F/S	Film Type : D7																		
Film Identification	No. Joint	Location	Welder Stamp	Type of Discontinuity												Film		Result	Remarks	
				Burn Through	Crack	Excessive Reinforcement	Tungsten Inclusions	Slag Inclusions	Incomplete fusion	Incomplete Penetration	Misalignment / Hi-Lo	Porosity	Root Concavity	Undercut	Film 4" X 10"	Film 4" X 15"				
PQR - PMC - ASME - I.1 - GS - 003																				
FADLI	6G	0-7														1	PASS			
		7-14															1	PASS		
		14-0															1	PASS		
TOTAL WELD : 1 JOINT										TOTAL OF FILM USED : 3										
Radiographer		QC/QA Inspector				Client Representative				Third Party										
Date :		Date :				Date :				Date : 26-08-2024										

Testing Report

Customer : PT. Pertamina Maintenance & Construction
Address : Graha PertaMC, Jl. Prof. DR. Soepomo SH
No. 55, Tebet Barat, Kec. Tebet,
Kota Jakarta Selatan, DKI Jakarta 12810
Reference Code : ASME Sect. IX : 2023
Test Method : ASTM E8/E8M-22
Object to be tested : (as received)

Job No : B24496
No. : HT/JKT/0824/0202
Total of Page : 1 of 2
Date of Received : 26 August 2024
Date of Tested : 30 August 2024
PO No : --
Project Name : --
Sample Marking : 8. PMC 13

One welded carbon steel pipe Joint test sample, size: Ø 6" SCH XXS (21.95 mm thickness), material spec: SA 106 Gr. B, WPS No. ~~WPS PMC ASME 1.1 GS-003~~, PQR No. PQR-PMC-ASME-1.1-GS-0023, welding process: GTAW+SMAW, welding position: 6G, filler metal: ER70S-6 (GTAW), E7018 (SMAW), for Welding Procedure Specification Test.

Type of Test : Reduced Section Tension Test
Environment Temp : 25.1°C

Test Specimen No	T1	T2
Measured Thickness (mm)	21.74	21.76
Measured Width (mm)	19.48	19.51
Effective Area (mm ²)	423.50	424.54
Ultimate Tensile Load (kN)	217.06	216.31
Ultimate Tensile Strength (N/mm ²)	513	510
Location of Failure	Base Metal	Base Metal
Type of Failure	Ductile	Ductile

Testing Machine Used : INSTRON 600 DX,
Model: 600 DX, S/N: S2513, Capacity : 600 kN
Remark : --
Test Conducted By : M. Anjar Rizki., ST
Witnessed By :

Verified By

Hadi Anwar S., ST
Coordinator Testing

Term & Conditions

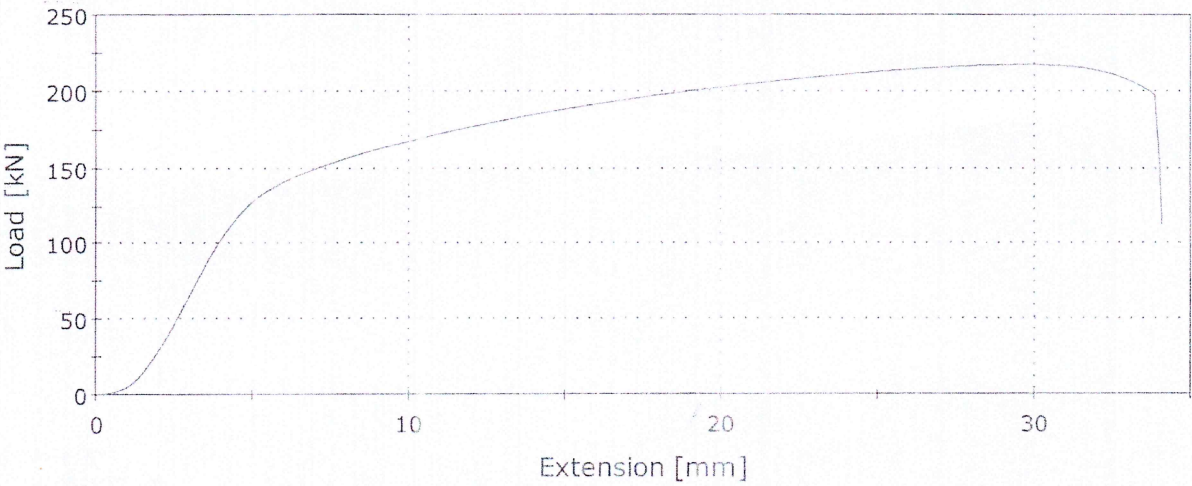
- The Report is prepared for the sole use of the Client and is prepared based upon the item submitted, the Services required by the Client and the conditions under which the Services are performed by PT. Hi-Test.
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PT. HI-TEST

Company
Report No
Sample/Stamp No

PT. Pertamina Maintenance & Construction
HT/JKT/0824/0202
8. PMC 13 - T2

Stress vs Strain



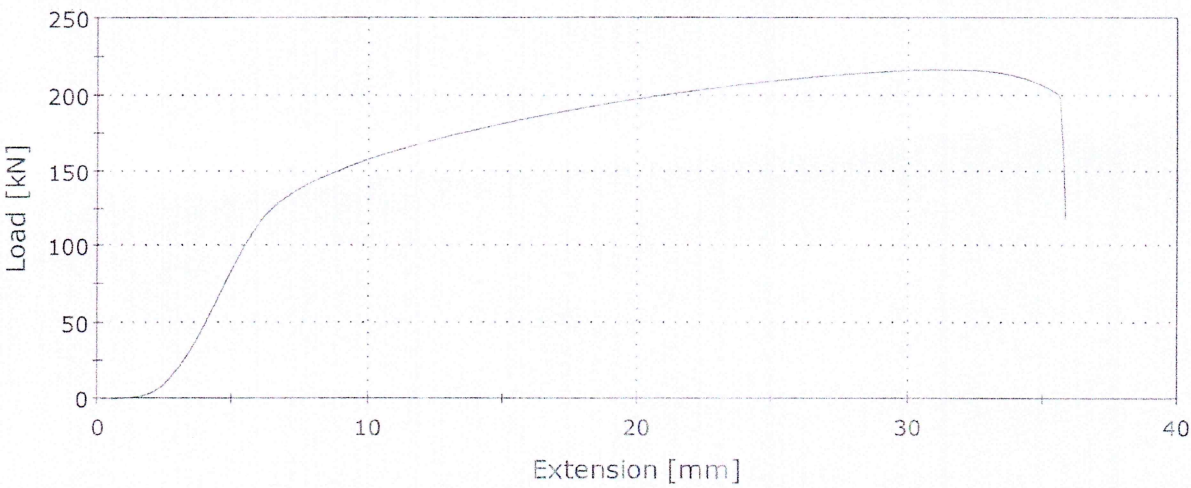
	Specimen label	Thickness [mm]	Width [mm]	Initial Area [mm^2]	Load at Tensile Strength [kN]	Tensile Stress [MPa]
1	8. PMC 13 - T2	21.74	19.48	423.50	217.06	513

PT. HI-TEST

Company
Report No
Sample/Stamp No

PT. Pertamina Maintenance & Construction
HT/JKT/0824/0202
8. PMC 13 - T1

Stress vs Strain



	Specimen label	Thickness [mm]	Width [mm]	Initial Area [mm^2]	Load at Tensile Strength [kN]	Tensile Stress [MPa]
1	8. PMC 13 - T1	21.76	19.51	424.54	216.31	510

Testing Report

Customer : PT. Pertamina Maintenance & Construction
Address : Graha PertaMC, Jl. Prof. DR. Soepomo SH
No. 55, Tebet Barat, Kec. Tebet,
Kota Jakarta Selatan, DKI Jakarta 12810
Reference Code : ASME Sect. IX : 2023
Test Method : ASTM E190-21
Object to be tested : (as received)

Job No : B24496
No. : HT/JKT/0824/0202
Total of Page : 2 of 2
Date of Received : 26 August 2024
Date of Tested : 30 August 2024
PO No : --
Project Name : --
Sample Marking : 8. PMC 13

One welded carbon steel pipe Joint test sample, size: Ø 6" SCH XXS (21.95 mm thickness), material spec: SA 106 Gr. B, ~~WPS No.~~
~~WPS-PMC-ASME-1.1-GS-003~~, PQR No. PQR-PMC-ASME-1.1-GS-003, welding process: GTAW+SMAW, welding position: 6G,
filler metal: ER70S-6 (GTAW), E7018 (SMAW), for Welding Procedure Specification Test.

Type of Test : Bending Test (Transverse Specimen)

Test Temperature : 25.1 °C
Test Weld Metal Thickness : 21.95 mm
Former Diameter : 40 mm
Shoulder Distance : 63 mm
Test Specimen Width : 10 mm
Bend Angle : 180 Degree

Specimen No	Type of Bend	Observation	Result
SB1	Side Bend	No Open Discontinuity Was Observed	Accepted
SB2	Side Bend	No Open Discontinuity Was Observed	Accepted
SB3	Side Bend	No Open Discontinuity Was Observed	Accepted
SB4	Side Bend	No Open Discontinuity Was Observed	Accepted

Testing Machine Used : MOTOR PUMP BENDING TEST MACHINE,
Model: Daventry England, S/N: DG 8881, Capacity: 700 Bar

Remark : --
Test Conducted By : Rifky Prasetyo Maulana, ST
Witnessed By : 

Verified By

Hadi Anwar S., ST
Coordinator Testing

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- The Report is prepared for the sole use of the Client and is prepared based upon the item submitted. The Services required by the Client and the conditions under which the Services are performed by PT. Hi-Test.
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SUMITOMO METAL INDUSTRIES, LTD.
WAKAYAMA STEEL WORKS (KAINAN)
260-100, FUNDO, KAINAN, JAPAN

INSPECTION CERTIFICATE

CERTIFICATE NO. :BYPP0076

PAGE : 1/1 DATE : 2012-03-22

CUSTOMER : PT. PARADISE PERKASA
ORDER NO. :
SHIPPER : SUMITOMO GASSAN CORPORATION 055 116 ACJ2030A 2P110460310
COMMODITY : SEAMLESS HOT FINISHED CARBON STEEL PIPE WITH BEVELED ENDS
STANDARD : API 5L GR. L290 PSL 1 / API 5L GR. L245 PSL 1 / ASTM A106M-08 / ASME 2010 (A11) SA-106M GR. B
ASTM A69M-07 / ASME 2010 (A11) SA-53M GR. B
SPECIFICATION :
MILL WORK NO. :BYPP0076 ← O.D. :168.3mm W.T. :21.95mm LENGTH:2000mm QUANTITY:160pcs. MASS:7605kg
HEAT NO. :J2K3135 ←
PRODUCTS PCS. :160

HEAT TREATMENT :AS ROLLED
FINISHING TEMPERATURE 780—1100°C

CHEMICAL COMPOSITION(S)

			C	Si	Mn	P	S	Co	Cr	Ni	Mo	Ti	V	Nb	Al	U	W	Bi	As	Se
		#1				#2	#2					#2		#2	#2					
SPEC. MIN.		R	-	10	29	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
MAX.		R	23	-	106	30	30	40	40	40	15	-	8	-	1	100	6	16	42	
HEAT NO.																				
J2K3135		L	20	24	98	13	6	3	8	3	1	1	0	10	0	13	1	1	38	
		P	19	24	97	12	5	3	6	3	1	1	0	8	0	13	1	1	37	
		P	19	24	97	12	6	3	3	3	1	1	0	8	0	13	1	1	37	
		P	19	24	99	12	6	3	8	3	1	2	0	9	0	13	1	1	37	
		P	19	24	99	13	6	3	5	3	1	1	0	9	0	13	1	1	37	

*1 RELIABLE & PRODUCT ANALYSIS L:LABILE ANALYSIS P:PRODUCT ANALYSIS *2:X1000 OTHER:X100 *U4:Cu+Ni+Cr+Mo+V
*W12:W *V1:Mo+V+Ti *CEQ:(C+Mn/6)+(Cr+Mo+V)/5+(Cu+Ni)/15

TENSILE TEST

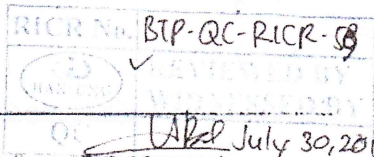
		#1	#2	YS	TS	EL
				#3	#3	%
SPEC. MIN.		L B		M 290	M 415	27
MAX.		L B		M -	M -	-
HEAT NO.						
J2K3135		L B		N 360	M 623	42
		L B		N 338	M 537	40

TYPE OF SPECIMEN:STRIP 25.4mm WIDTH *1 SAMPLING DIRECTION L:LONGITUDINAL *2 SAMPLING POSITION B:BASE METAL
*3 UNIT N:MPa GAUGE LENGTH:50.0mm KIND OF YS:0.5% EXTENSION UNDER LOAD

FLATTENING TEST: ACCEPTABLE
VISUAL & DIMENSIONS: ACCEPTABLE
HYDROSTATIC TEST 20.5MPa: ACCEPTABLE
EN 10204 3.1
NACE MR0175/ISO 15156-2/2003 & NACE 0103 PART 2.1

Item No. : 1
CERTIFIED TRUE COPY
Signature : PT. PARADISE PERKASA

WE HEREBY CERTIFY THAT THE MATERIAL HEREIN DESCRIBED HAS BEEN MANUFACTURED, SAMPLED, TESTED, AND INSPECTED IN ACCORDANCE WITH ABOVE STANDARD AND SPECIFICATION AND SATISFIES THE REQUIREMENTS.



T. Kondo
MANAGER, QUALITY ASSURANCE SECTION

(An Important Message To Our Customers)
This certification is intended only for products listed. Modification to, or unauthorized use of this certification is strictly prohibited. Offences may be regarded as forgery of documents and be subject to criminal prosecution. If you have any questions on this certification, you can contact us by facsimile or e-mail as shown below.
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1071 62

PT ALLALLOY CAHAYA DYNAWELD
JL. PANTAI INDAH SELATAN KOMP.
KAMAL MUARA, PENJARINGAN
14470 Jakarta Utara
Indonesia

Inspection certificate 3.1

as per : EN 10204

No. : 2023-2047018962-10-A800061-025

Rev. 0

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PO no.	ACD/DN-N/23/04/0423	of	27.04.2023
Order no.	1042007091		
Delivery note/pos./split	2047018962/000000/000010	of	26.06.2023
Product	GTAW rod/wire		
Trade name	BOHLER S EMK 6		
Standard designation	EN ISO 636-A : W 42 5 3Si1 AWS A5.18/SFA 5.18 : ER70S-6		
Dimension	2,4 x 1000mm		118955
Lot no.	A800061		86356
Quantity	120,0 KG		JU10B00W 0425

Chemical composition in % of the product

C	Si	Mn	P	S	Cr	Mo	Ni	V	Cu	Al	Ti+Zr			
0,07	0,86	1,45	0,023	0,014	0,03	0,010	< 0,01	0,002	0,02	< 0,001	< 0,01			

Mechanical properties

ASME II C; Sch F

Tensile test				according to : ASTM E8			
Specimen preparation				according to : AWS B4.0			
T	ReL / Rp 0,2	Rp 1,0	Rm	A (Lo = 4d)	Z	WBH	Remarks
	MPa	MPa	MPa	%	%	PWHT	
25°C	430		540	40		AW	Typical values
Impact test				according to : ASTM E23			
Specimen preparation				according to : AWS B4.0			
T	Impact energy	Average	Lateral expansion	Shear fracture		WBH	Remarks
	KV / J	KV / J	mm	%		PWHT	
-50°C		90				AW	Typical values

The product BOHLER S EMK 6 meets the requirements of the filler metal specification ASME sec II, part C, AWS A5.18/SFA 5.18 : ER70S-6 when tested in accordance with that specification.

Complying the corrosion testing requirements as per NACE MR 0175 standard TM 01-77-96 (SSC) and 02-84-96 (HIC).

Town
Bekasi

Date
11.07.2023

This certificate was issued by DP-equipment and does not require signature.

Authorized representative
Miftah Farid

PT ALLALLOY CAHAYA DYNAWELD
JL. PANTAI INDAH SELATAN KOMP.
KAMAL MUARA, PENJARINGAN
14470 Jakarta Utara
Indonesia

Inspection certificate 3.1

as per : EN 10204

No. : 2022-2047015416-10-22289-025

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PO no.	ACD/JK-N/21/12/1132	of	15.12.2021
Order no.	1042005811		
Delivery note/pos./split	2047015416/000000/000010	of	06.01.2022
Product	covered electrode		
Trade name	BOHLER FOX S EV 50		
Standard designation	ISO 2560-B : E 49 18 A U H5		
	ISO 2560-A : E 42 3 B 42 H 5		
	AWS A5.1/SFA 5.1 : E7018 H4R		
Dimension	SMAW 3,25 x 350 mm		
Lot no.	22289		
Quantity	10000,0 KG		

118955
82555
JUP8000E
0425

Chemical composition in % of the weld metal

C	Si	Mn	P	S	Cr	Mo	Ni	V	Cu	Nb				
0,08	0,48	1,38	0,020	0,005	0,024	0,003	0,009	0,003	0,008	0,001				

Mechanical properties

Tensile test				according to : ASTM E8			
Specimen preparation				according to : AWS B4.0			
T	ReL / Rp 0,2	Rp 1,0	Rm	A (Lo = 4d)	Z	WBH	Remarks
	MPa	MPa	MPa	%	%	PWHT	
25°C	490		560	30		AW	
Impact test				according to : ASTM E23			
Specimen preparation				according to : AWS B4.0			
T	Impact energy	Average	Lateral expansion	Shear fracture	WBH	Remarks	
	KV / J	KV / J	mm	%	PWHT		
-30°C		55			AW		

The product BOHLER FOX S EV 50 meets the requirements of the filler metal specification ASME sec II, part C, AWS A5.1/SFA 5.1 : E7018 H4R when tested in accordance with that specification.

Complying the corrosion testing requirements as per NACE MR 0175 standard TM 01-77-96 (SSC) and 02-84-96 (HIC).

Produced according to AWS A5.01, class C1.

Level of testing schedule F.

Town
Bekasi

Date
13.01.2022

This certificate was issued by DP-equipment and does not require signature.

Authorized representative
Miftah Farid

PT ALLALLOY CAHAYA DYNAWELD
JL.PANTAI INDAH SELATAN KOMP.
PENJARINGAN, KOTA ADM.
14470 JAKARTA UTARA, DKI JAKARTA,
Indonesia

Inspection certificate 3.1

as per : EN 10204

No. : 2023-2047019662-10-A100456-025

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PO no.	ACD/BS-N/23/08/0819	of	22.08.2023
Order no.	1042007417		
Delivery note/pos./split	2047019662/000000/000010	of	08.09.2023
Product	covered electrode		
Trade name	BOHLER FOX S EV 50		
Standard designation	ISO 2560-B : E 49 18 A U H5 ISO 2560-A : E 42 3 B 42 H5 AWS A5.1M : E4918H4R AWS A5.1 : E7018 H4R		
Dimension	4,0 x 350 mm dry system		
Lot no.	A100456		
Quantity	48,0 KG		

Chemical composition in % of the weld metal

C	Si	Mn	P	S	Cr	Mo	Ni	V	Cu	Nb				
0,08	0,31	1,07	0,024	0,006	0,034	0,001	0,016	0,003	0,024	0,001				

Mechanical properties**ASME II C; Sch F**

Tensile test				according to : ASTM E8			
Specimen preparation				according to : AWS B4.0			
T	ReL / Rp 0,2	Rp 1,0	Rm	A (Lo = 4d)	Z	WBH	Remarks
	MPa	MPa	MPa	%	%	PWHT	
25°C	490		560	30		AW	
Impact test				according to : ASTM E23			
Specimen preparation				according to : AWS B4.0			
T	Impact energy	Average	Lateral expansion	Shear fracture	WBH	Remarks	
	KV / J	KV / J	mm	%	PWHT		
-30°C		55			AW		

The product BOHLER FOX S EV 50 meets the requirements of the filler metal specification ASME sec II, part C, AWS A5.1 : E7018 H4R when tested in accordance with that specification.

Complying the corrosion testing requirements as per NACE MR 0175 standard TM 01-77-96 (SSC) and 02-84-96 (HIC).

Town
Bekasi

Date
14.09.2023

This certificate was issued by DP-equipment and does not require signature.

Authorized representative
Miftah Farid